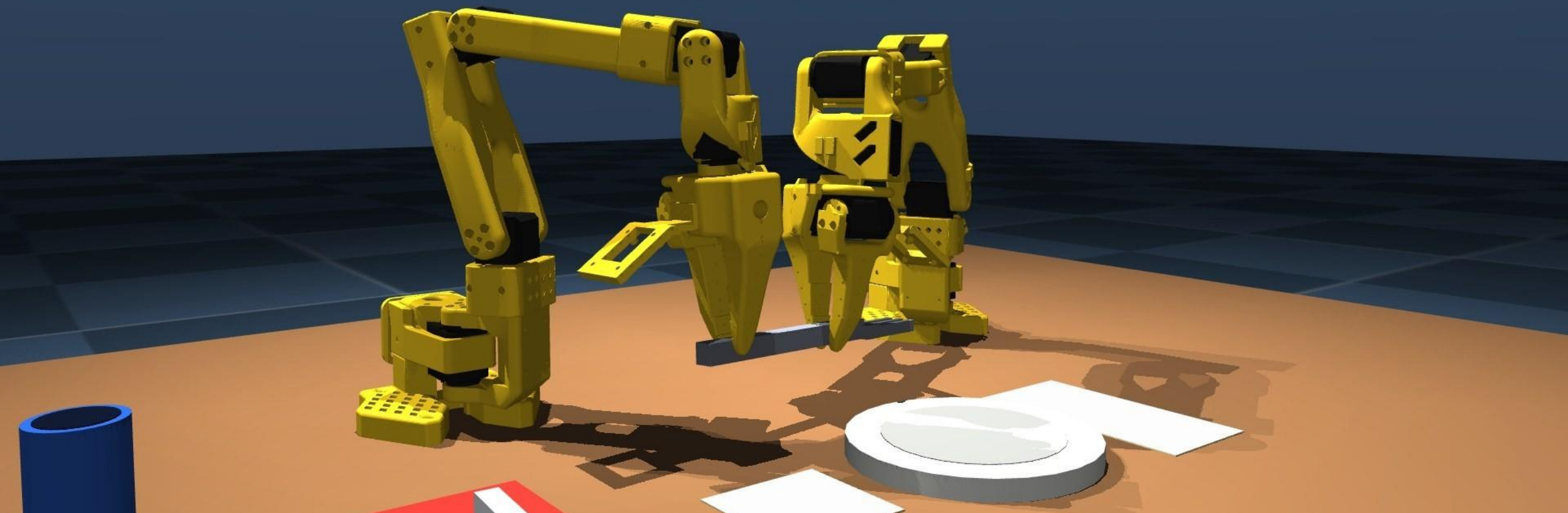


RescueHands AI

Two SO-101 arms set a dinner table — and recover when a fork drops

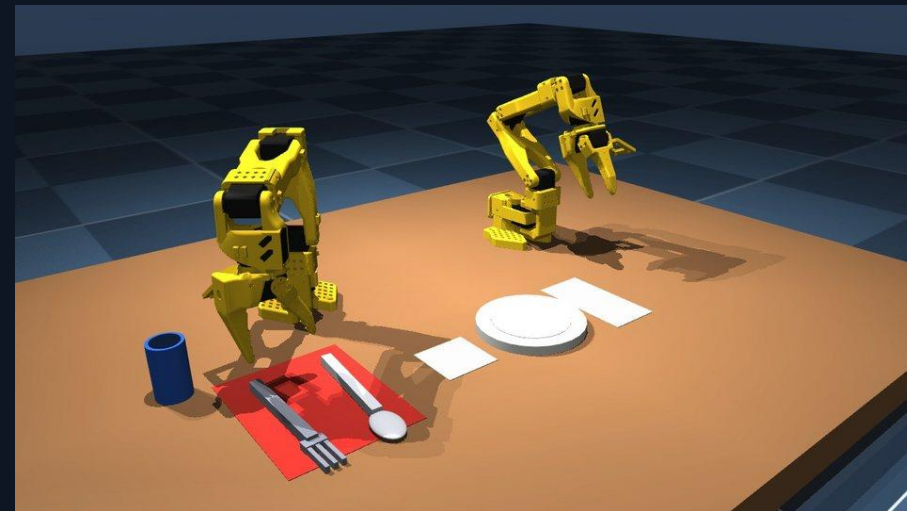
SmolVLA · MuJoCo · physics-aware supervisor · OpenVINO on Intel CPU + iGPU



THE PROBLEM

A policy acts even after reality went wrong

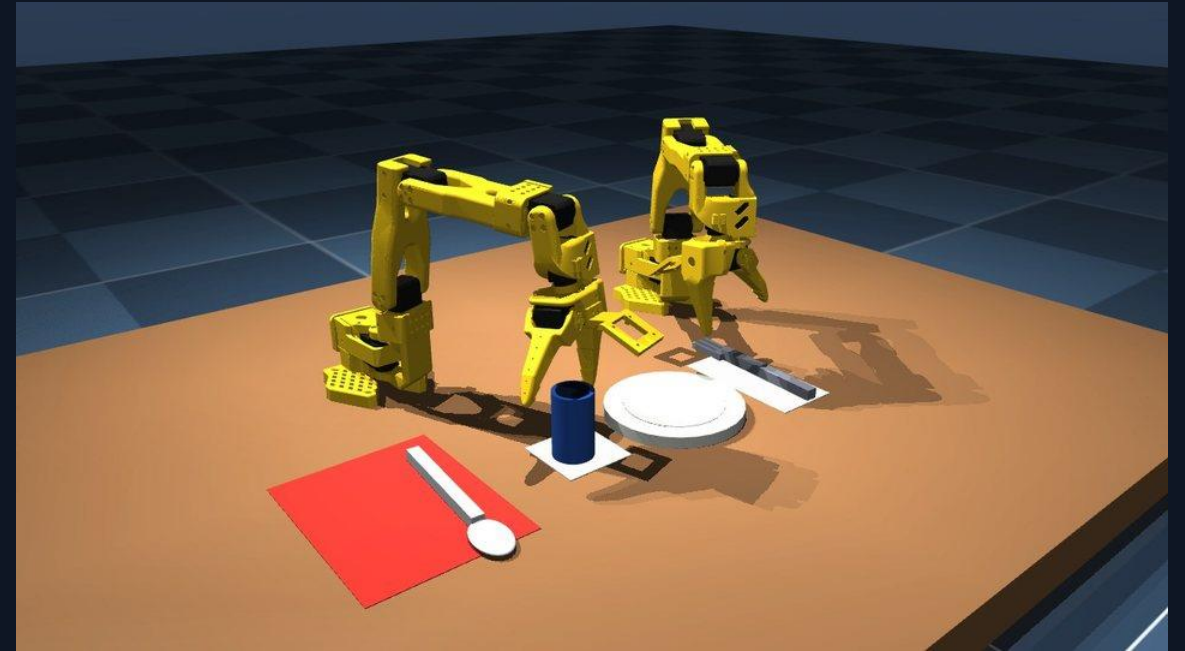
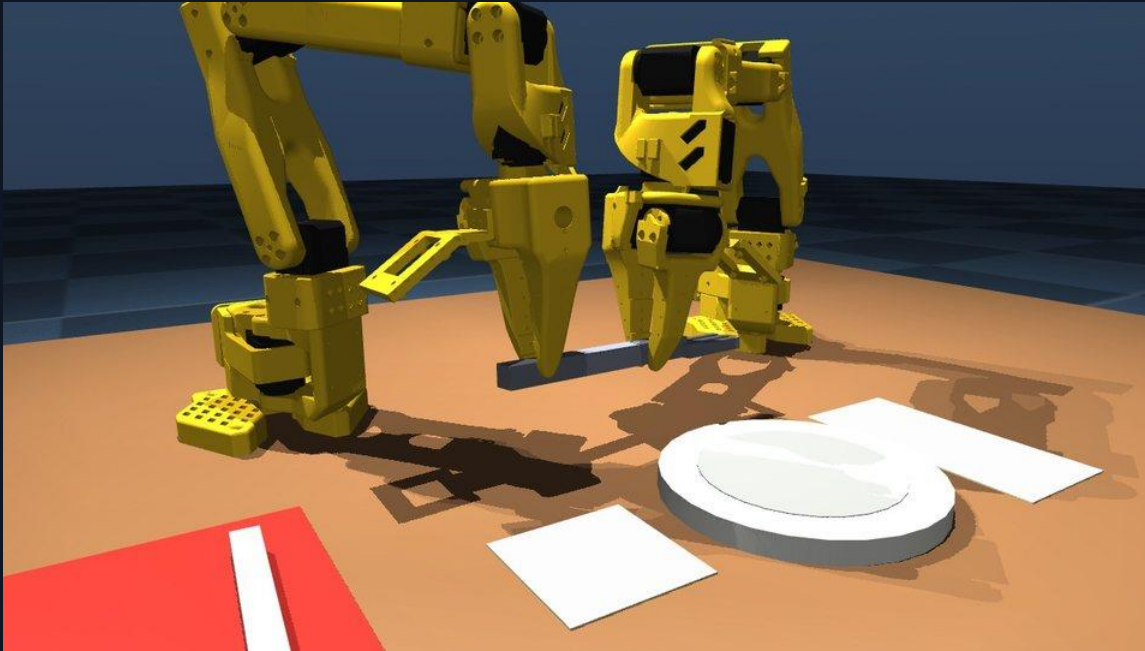
- Vision-language-action models predict the next move; they do not check the last one.
- With two arms, one slip by one hand ruins the other hand's work.
- Learned policies drift into states their demonstrations never showed.
- So: never trust an action because a model produced it — check the physics, then recover.



A real drop: the gripper is forced open, the fork falls, both arms back off and retry.

THE TASK

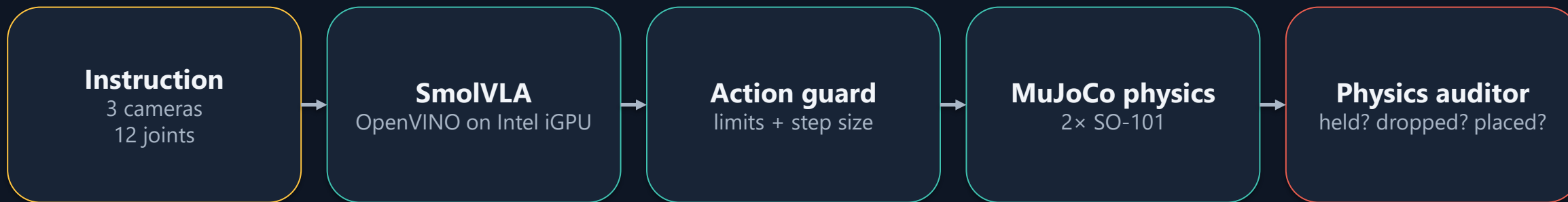
Set a dinner place with two SO-101 arms



- "Hand the spoon over to the left arm, then place the cup next to the plate." — the words pick fork or spoon
- Right arm picks → in-air hand-off → left arm places the utensil → right arm places the cup
- Every seed changes positions, sizes, mass, friction, lighting and table colour; contact grasps only, no welding

ARCHITECTURE

The VLA proposes; physics decides



Policy sees

Camera images, joint positions and the instruction. Nothing else — no object poses, no contacts.

Supervisor sees

Simulator ground truth: which jaws hold what, support, zones, arm-to-arm contact. Used only to judge and recover.

IK only in the teacher

A scripted IK teacher makes demonstrations and the baseline. The deployed loop has no IK: SmoIVLA outputs the 12 joint targets.

LEARNED POLICY

SmolVLA, taught to get back on track

	v1	v2
Episodes	101	569
Frames (20 fps)	57,173	336,729
Clean demonstrations	101	293
Messy starts (arms nudged, items moved)	—	194
Real drop → recovery → finish	—	82
Training	12,000 steps from SmolVLA base	6,000 steps from v1, lr 5e-5

- Why v2: v1 copied the teacher closely (open-loop error ≈ 0.01 rad) but only 1 of 9 completed live episodes succeeded — live runs reach states the clean demos never showed.
- Trained on free Kaggle T4 GPUs; camera images rendered on the GPU (229 s → 14 s per episode after an EGL fix).

Physics decides success — not the policy

Supervisor

Drop or stalled grasp → open both hands, return to a safe pose, let the policy re-plan. At most 2 retries. The retreat itself is watched: an item leaving the table ends the episode.

Task success (10 steps stable)

Cup and utensil in their zones, released, supported, cup upright ($\leq 15^\circ$), settled, spare utensil still near its start at the end, and an ordered IN-AIR hand-off: right alone → both → left alone, with at most 5 steps of contact flicker.

Checked, not assumed

Model contracts pin joint order, cameras, rate and file hashes. Every run writes a manifest (code revision, arguments, versions). An independent audit's findings were reproduced, fixed and re-measured.

RESULTS

Results on 10 randomized seeds

Policy	Supervisor	Gripper fault	Success
Scripted teacher (baseline)	on	none	8/10
Scripted teacher (baseline)	off	yes	2/10
Scripted teacher (baseline)	on	yes	7/10
SmoIVLA v2 · OpenVINO iGPU	off	none	<y>/10
SmoIVLA v2 · OpenVINO iGPU	on	none	<x>/10
SmoIVLA v2 · OpenVINO iGPU	off	yes	<w>/10
SmoIVLA v2 · OpenVINO iGPU	on	yes	<z>/10

Pairs change one thing: same seeds and fault, supervisor on vs off. For the teacher it turns 2/10 into 7/10. Every number comes from committed result files with run manifests.

Running the policy on an Intel laptop

Backend (Intel Core i5-6300U + HD Graphics 520)	Time per 50-action chunk	Max action diff vs FP32
PyTorch, CPU (baseline)	$\langle b \rangle$ s	$\langle d \rangle$ rad
OpenVINO FP32, CPU	$\langle e \rangle$ s	0 (reference)
OpenVINO FP16, iGPU	$\langle a \rangle$ s	$\langle d \rangle$ rad
OpenVINO INT8 weights (NNCF)	$\langle f \rangle$ s	$\langle g \rangle$ rad

- iGPU speed-up over PyTorch CPU: $\langle c \rangle \times$, with its accuracy cost reported next to it
- Exported with Intel Physical AI Studio; organizers allowed non-Core-Ultra Intel hardware
- Load time and memory are measured inside the same timer for every backend

What we claim, and what we do not

Limits

- Collision checks cover arm-to-arm contact only.
- Randomized item friction does not reach the jaw contacts (measured), so slippery grasps are not tested.
- Simulation pauses during inference: not real-time 20 Hz control.
- Evaluation seeds guided teacher debugging, so they are not a pristine unseen set.

Reproduce it

- Pinned MuJoCo scene and SO-101 asset revision
- Data generation, training, export, evaluation, benchmark scripts
- Public dataset ABDHAM/rescuehands_table_v2 and model ABDHAM/smolvla_rescuehands_v2
- 99 unit and physics tests · per-run manifests · dataset provenance logs

THANK YOU

RescueHands AI

A VLA should not be trusted just because it produced an action. Check what physically happened — and recover.

github.com/Hamza-Atiq/rescuehandsAI

huggingface.co/ABDHAM/smolvla_rescuehands_v2

huggingface.co/datasets/ABDHAM/rescuehands_table_v2